

# SIMATS ENGINEERING

**SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES  
CHENNAI – 602105**

**Department of Computer Science and Engineering**

## ASSESSMENT TOOL 2 – APPLICATION BASED QUESTIONS

|               |                                      |
|---------------|--------------------------------------|
| Course Code:  | DSA0501                              |
| Course Title: | Query Processing using Data Analysis |
| CO Assessed:  | CO1                                  |
| Assessment:   | Tool 2 – Application Based Questions |
| Weightage:    | _____                                |
| Total Marks:  | 20 Marks                             |

CO1 Statement: \_\_\_\_\_

Student Name: U.Ananya      Reg. No.: 19242438

Section/Batch: AI&DS / 2024 Batch

Date of Submission: 27-07-2026

Faculty In-charge: \_\_\_\_\_ Submission Mode: Individual

### Code of Conduct

I U.Ananya certify that this submission is my original work and have adhered to all academic integrity guidelines.

Signature:U.Ananya

| Criteria                                | Marks |
|-----------------------------------------|-------|
| Understanding of the Problem            | 4     |
| Application of Concepts                 | 4     |
| Program Design / Algorithm              | 4     |
| Correctness of Solution & Output        | 4     |
| Explanation / Interpretation of Results | 4     |
| Total                                   | 20    |

## CO1 – AT2 – Application Based Questions

### Sample Questions

#### 1. Library Database Preparation (10 Marks)

##### Library Database Preparation using JSON File

##### Python Program code:

```
import json

# Read JSON file

with open("books.json", "r") as file:

    books = json.load(file)

cleaned_books = []

# Validate required fields

for book in books:

    if (book.get("Book ID") and

        book.get("Title") and

        book.get("Author") and

        book.get("Publisher") and

        book.get("Year")):

        cleaned_books.append(book)

    else:

        print("Invalid Record:", book)

# Save cleaned data

with open("cleaned_books.json", "w") as file:

    json.dump(cleaned_books, file, indent=4)

print("Data cleaned successfully and saved in cleaned_books.json")
```

##### Output:

```

>>> = RESTART: C:/Users/uppal/AppData/Local/Programs/Python/Python313/library database.py
Invalid Record: {'Book ID': 'B102', 'Title': 'Data Science', 'Author': '', 'Publisher': 'XYZ Publications', 'Year': 2022}
Data cleaned successfully and saved in cleaned_books.json
>>> ===== RESTART: C:/Users/uppal/AppData/Local/Programs/Python/Python313/library database.py =====
Invalid Record: {'Book ID': 'B102', 'Title': 'Data Science', 'Author': '', 'Publisher': 'XYZ Publications', 'Year': 2022}
Data cleaned successfully and saved in cleaned_books.json

Cleaned Dataset:
[
  {
    "Book ID": "B101",
    "Title": "Python Basics",
    "Author": "John",
    "Publisher": "ABC Publications",
    "Year": 2023
  },
  {
    "Book ID": "B103",
    "Title": "Machine Learning",
    "Author": "David",
    "Publisher": "Tech Books",
    "Year": 2021
  }
]
>>> |

```

## 2. Banking Transaction Analysis (10 Marks)

### Python Program Code:

```

import pandas as pd

# Read CSV file

df = pd.read_csv("transactions.csv")

# Remove duplicate Transaction IDs

df = df.drop_duplicates(subset="Transaction ID")

# Remove rows with blank Account Number

df = df[df["Account Number"].notna()]

# Standardize Transaction Type

df["Transaction Type"] = df["Transaction Type"].str.upper()

# Keep only valid transaction types

valid = ["DEPOSIT", "WITHDRAWAL", "TRANSFER"]

invalid = df[~df["Transaction Type"].isin(valid)]

print("Invalid Transactions:")

print(invalid)

# Remove invalid transactions

df = df[df["Transaction Type"].isin(valid)]

# Save cleaned data

df.to_csv("cleaned_transactions.csv", index=False)

print("Cleaned data saved in cleaned_transactions.csv")

```

### Output:

```
===== RESTART: C:/Users/uppal/AppData/Local/Programs/Python/Python313/transactions.py =====
transactions.csv created successfully
Invalid Transactions:
  Transaction ID  Account Number  Transaction Type  Amount
5             T105         10005.0          PAYMENT    2500
Cleaned data saved in cleaned_transactions.csv
```

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